

27-11-2023



1

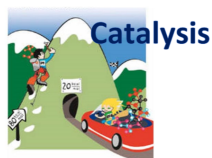


2

Objectives

- Catalysis/ Catalysts
- Steps involved
- Development of rate law
- Design of catalytic reactors

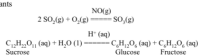
3



4

Catalysis

- Catalyst** is a substance that increases the *rate of the reaction* at which a chemical system approaches equilibrium, without being substantially consumed in the process.
- The selection of a catalyst to promote a reaction is *not well understood*; therefore, in practice extensive trial and error may be needed.
- Homogeneous catalyst**: If the catalyst is present in the same phase as the reactants

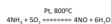


BITS Pilani, Pilani Campus

5

Heterogeneous catalysis

- If the catalyst is present in a different phase than the reactants



BITS Pilani, Pilani Campus

6

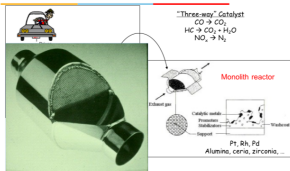
Heterogeneous reactions

- Haber-Bosch process**
 - $\text{N}_2 + 3 \text{H}_2 \rightarrow 2 \text{NH}_3$
 - Fe/Ru catalysts, high pressure and temperature
 - Critical for fertiliser and nitric acid production
- Fischer-Tropsch chemistry**
 - $n \text{CO} + 2n \text{H}_2 \rightarrow [\text{CH}_2]_n + n \text{H}_2\text{O}$, syn gas to liquid fuels
 - Fe/Co catalysts
- Fluidised catalytic cracking**
 - High MW petroleum $\rightarrow$ low MW fuels, like gasoline
 - Zeolite catalysts, high temperature combustor
- Automotive three-way catalysts**
 - $\text{NO}_x/\text{CO}/\text{HC} \rightarrow \text{H}_2\text{O}/\text{CO}_2/\text{H}_2\text{O}$
 - Pt/Rh/Pd supported on ceria/alumina
 - Makes exhaust 99% cleaner

BITS Pilani, Pilani Campus

7

Automotive Emissions Control System



BITS Pilani, Pilani Campus

8

Three-way Catalyst



BTS Plant, Plant Campus

9

Applications

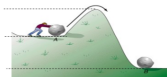
- Almost all chemical industries have one or more steps employing catalysts
Petroleum, energy sector, fertiliser, pharmaceutical, fine chemicals ...
- **Advantages of catalytic processes**
 - Achieving better process economics and productivity
 - Increase reaction rates - fast
 - Simplify the reaction steps - low investment cost
 - Carry out reaction under mild conditions (e.g. low T, P) - low energy consumption
 - Reducing wastes
 - Improving selectivity toward desired products - less raw materials required, less unwanted wastes
 - Replacing harmful/basic materials with readily available ones
 - Producing certain products that may not be possible without catalysts
 - Having better control of process (safety, flexible etc.)
 - Encouraging application and advancement of new technologies and materials
 - And many more ...

BTS Plant, Plant Campus

10

Activation energy

- **Activation Energy** : The energy required to overcome the reaction barrier
- The Activation Energy (E_a) determines how fast a reaction occurs, the higher Activation barrier, the slower the reaction rate. *The lower the Activation barrier, the faster the reaction*

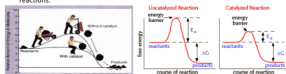


BTS Plant, Plant Campus

13

Activation energy

- Catalyst lowers the activation energy for both forward and reverse reactions.



- This means , the catalyst **changes the reaction path** by lowering its activation energy and consequently the catalyst **increases the rate of reaction**.

BTS Plant, Plant Campus

14

Action of the catalyst

- It is important to remember that the use of catalyst **DONES NOT** vary ΔG° & K_{eq} values of the reaction concerned, it **merely** changes the **RNCE** of the process.
 - Whether a reaction can proceed or not and to what extent a reaction can proceed is solely determined by the reaction thermodynamics, which is governed by the values of ΔG° & K_{eq} **NOT** by the presence of catalyst.
 - The reaction thermodynamics provide the driving force for a rxn; the presence of catalysts changes the way how driving force acts on that process.
- $$\text{CH}_4(g) + \text{CO}_2(g) \rightleftharpoons 2\text{CO}(g) + 2\text{H}_2(g) \quad \Delta G^\circ_{298} = 151 \text{ kJ/mol} \quad (1800^\circ\text{C})$$
- $$\Delta G^\circ_{298} = 18 \text{ kJ/mol} \quad (700^\circ\text{C})$$
- At 1800°C , $\Delta G^\circ_{298} = 151 \text{ kJ/mol} > 0$. There is no thermodynamic driving force, the reaction won't proceed with or without a catalyst.
 - At 700°C , $\Delta G^\circ_{298} = 18 \text{ kJ/mol} < 0$. The thermodynamic driving force is there. **However, simply putting CH_4 and CO_2 together is a reaction blocked issue that will stall.** Without a proper catalyst heating the mixture in reactor results in no conversion of CH_4 and CO_2 at all. When $\text{Pt}(\text{O}_2)$ or $\text{Ni}(\text{O}_2)$ is present in the reactor at the same temperature, equilibrium conversion can be achieved (>100%).

BTE Plant, Plant Campus

15

Solid Catalysts

Catalyst composition

- Active phase**
 - Where the reaction occurs (mostly metal/metal oxide)
- Promoter**
 - Textual promoter (e.g. Al - Fe for NH_3 production)
 - Electric or Structural modifier
 - Poison-resistant promoters
- Support / carrier**
 - Increase mechanical strength
 - Increase surface area (98% surface area is supplied within the porous structure)
 - may or may not be catalytically active
- Gauze / foil:** External active sites are enough to get the faster rates



BTE Plant, Plant Campus

16

Solid support / carrier materials

- Some common Alumina**
 - Inexpensive
 - Surface area: $1 \sim 700 \text{ m}^2/\text{g}$
 - Acidic/ neutral
- Silica**
 - Inexpensive
 - Surface area: $100 \sim 800 \text{ m}^2/\text{g}$
 - Acidic
- Zirconite**
 - mixture of alumina and silica,
 - often exchanged metal ion present
 - shape selective
 - Acidic
- Other supports**
 - Active carbon (S.A. up to $3000 \text{ m}^2/\text{g}$)
 - Titanium (S.A. $10 \sim 50 \text{ m}^2/\text{g}$)
 - Zirconia (S.A. $10 \sim 100 \text{ m}^2/\text{g}$)
 - Magnesia (S.A. $10 \text{ m}^2/\text{g}$)
 - Lanthana (S.A. $10 \text{ m}^2/\text{g}$)



BTE Plant, Plant Campus

17

Periodic Table of the Elements

Transition metals and their compounds function as catalysts either because of their ability to change oxidation state or, in the case of the metals, to adsorb other substances on to their surface and activate them in the process.

BTE Plant, Plant Campus

18

Physical properties

Support/Catalyst	BET area (m ² /g)	Pore Vol.	Pore Diam. (nm)
Activated Carbon	500-1500	0.6-0.8	0.6-2
Zeolites (Molecular Sieves)	500-1000	0.5-0.8	0.4-1.8
Silica Gels	200-600	0.40	3-20
Activated Clays	150-225	0.4-0.52	20
Activated Al ₂ O ₃	100-300	0.4-0.5	6-40
Kieselguhr ("Celite 296")	4.2	1.14	2,200

BET: Brunauer, Emmett and Teller

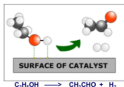
BITS Pilani, Pilani Campus

20

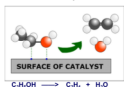
Specificity choice of catalyst can influence the products

- In some cases the Ethanol undergoes different reactions depending on the metal used as the catalyst.
- The distance between active sites and their similarity with the length of bonds
- determines the method of adsorption and affects which bonds are weakened.

Copper Dehydrogenation (oxidation)



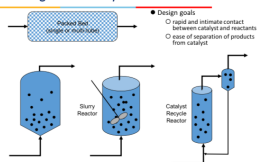
Alumina Dehydration



BITS Pilani, Pilani Campus

21

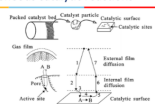
Heterogeneous catalytic reactions



BITS Pilani, Pilani Campus

22

Heterogeneous catalytic reactions



BITS Pilani, Pilani Campus

23

Intrinsic Catalytic reaction

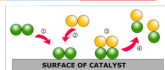
- **Catalytic action** takes place at **active sites** on the surface of a solid
 - gases are adsorbed onto the surface
 - they form bonds with metal atoms
- Catalysis is thought to work in three stages...

Adsorption
Reaction
Desorption

24

BTS Pinar, Pinar Campus

Intrinsic reaction steps

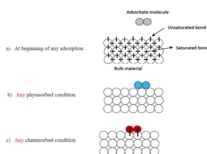


- Adsorption (STEP 1)**
The incoming species lands on an active site and forms bonds with the catalyst. It may use some of the bonding electrons in the molecules thus weakening them and making a subsequent reaction easier.
- Reaction (STEPS 2 and 3)**
Adsorbed gases may be held on the surface in just the right orientation for a reaction to occur. This increases the chances of favourable collisions taking place.
- Desorption (STEP 4)**
There is a re-arrangement of electrons and the products are then released from the active sites

25

BTS Pinar, Pinar Campus

Adsorption



26

BTS Pinar, Pinar Campus

Adsorption On Solid Surface

- **Adsorption** is a process in which molecules from gas (or liquid) phase land on, interact with and attach to solid surfaces.
- The reverse process of adsorption, i.e. the process in which adsorbed molecules escape from solid surfaces, is called **Desorption**.
- Molecules can attach to surfaces in two different ways because of the different forces involved. These are **Physisorption** (Physical adsorption) & **Chemisorption** (Chemical adsorption)

	Physisorption	Chemisorption
force	van de Waal	chemical bond
number of adsorbed layers	multi	only one layer
adsorption heat	low (10-40 kJ/mol)	high (> 40 kJ/mol)
selectivity	low	high
temperature to occur	low	high

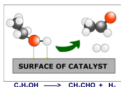
27

BTS Pinar, Pinar Campus

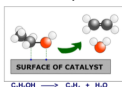
Specificity *choice of catalyst can influence the products*

- In some cases the Ethanol undergoes different reactions depending on the metal used as the catalyst.
- The distance between active sites and their similarity with the length of bonds
- determines the method of adsorption and affects which bonds are weakened.

Copper Dehydrogenation (oxidation)



Alumina Dehydration



BTS Planet, Planet Campus

28

Adsorption mechanisms

Langmuir-Hinshelwood mechanisms:

1. Adsorption from the gas-phase
2. Reactions between adsorbed molecules
3. Dissociation of molecules at the surface
4. Desorption to the gas-phase

Two Questions:

- Is the reaction has a Langmuir-Hinshelwood mechanism?
- What is the precise nature of the reaction steps?

Cannot be solved

without experimental or computational studies

BTS Planet, Planet Campus

29

Langmuir-Hinshelwood mechanisms

The Reaction $\text{A}_2 + 2\text{B} = 2\text{AB}$

may have the following mechanism



BTS Planet, Planet Campus

30

Eley-Rideal mechanism

- Adsorption from the gas-phase
- Dissociation of molecules at the surface
- Reactions between adsorbed molecules
- Reactions between gas and adsorbed molecules
- Desorption to the gas-phase

The 4th step cannot occur in a Langmuir-Hinshelwood mechanism

BTS Planet, Planet Campus

31

Eley-Rideal mechanism

Example



may have the following Eley-Rideal mechanism



where the last step is the direct reaction between the adsorbed molecule A^* and the gas-molecule B .

32

BTS Ploem, Ploem Campus

Eley-Rideal or Langmuir-Hinshelwood?

For the Eley-Rideal mechanism:

- The rate will increase with increasing coverage until the surface is completely covered by A^* .



For the Langmuir-Hinshelwood mechanism:

- The rate will go through a maximum and end up at zero, when the surface is completely covered by A^* .

This happens because the step $B + * \rightarrow B^*$ cannot proceed when A^* blocks all sites.

The trick is that the step $B + * \rightarrow B^*$

requires a free site

33

BTS Ploem, Ploem Campus

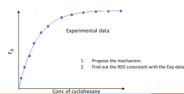
Intrinsic kinetics- Algorithm

- Rate equations for all the steps
- Assume one of the steps is RDS – intrinsically slowest.
- $r_0 = r_{RDS}$
- Rest all are in equilibrium, $r = 0$
- Express the surface adsorbed species conc. in terms of bulk species
- Site balance
- Substitute adsorbed species eqns and C_p in r_{RDS}

34

BTS Ploem, Ploem Campus

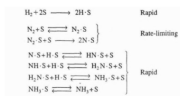
Cyclohexane dehydrogenation



35

BTS Ploem, Ploem Campus

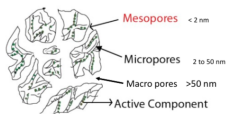
Ammonia synthesis



36

BTS Plant, Plant Campus

Pore network



38

BTS Plant, Plant Campus

Diffusion and Reaction in Spherical Catalyst Pellets

- Active sites are present inside the pore
- The reactant should diffuse first & then reaction takes place at the active sites.
- If molecule is moving (tortuous path) inside the pore, it experiences the resistance due to the very small pore sizes. (micro, meso, macro)
- Pores are in random size, except for zeolites.



39

BTS Plant, Plant Campus

Diffusion and Reaction in Spherical Catalyst Pellets

- Effective diffusivity, $D_e = \frac{D_{AB} \phi_e}{\tau}$

$$\tau = (\text{tortuosity}) = \frac{\text{Actual distance a molecule travels between two points}}{\text{Shortest distance between those two points}}$$



$$\begin{aligned}
 \beta &= \frac{\text{area } A_2}{\text{area } A_1} \\
 \sigma_0 &= 1/\beta
 \end{aligned}$$



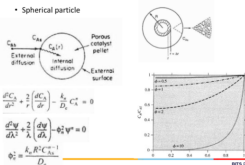
$$\tau = \frac{2L}{\sqrt{2}L} = \sqrt{2} = 1.414$$

40

BTS Plant, Plant Campus

Internal resistance

- Spherical particle



42

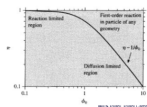
Internal effectiveness factor

- It tells how effective is the catalyst
- Works exactly opposite to Thiele modulus

$\eta = \frac{\text{Actual overall rate of reaction}}{\text{Rate of reaction that would result if entire interior surface were exposed to the external pellet surface conditions } C_{A,s}, T_s}$

$$\eta = \frac{-r_p^s}{-r_p^s} = \frac{-r_p^s}{-r_p^s} = \frac{-r_p^s}{-r_p^s}$$

$$\eta = \frac{3}{\phi_1} (\phi_1 \coth \phi_1 - 1)$$



43

Different geometries

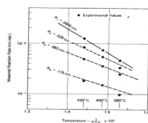
	Slab ($ae = x/R_p$)	Cylinder ^(2D) ($ae = r/R_p$)	Sphere ($ae = r/R_p$)
ϕ	$r_p \sqrt{\frac{k_p}{D_e}}$	$R_p \sqrt{\frac{k_p}{D_e}}$	$R_p \sqrt{\frac{k_p}{D_e}}$
$\psi = C_A/C_{A,s}$	$\cosh(\phi ae)$ $\cosh(\phi)$	$I_0(\phi ae)$ $I_0(\phi)$	$\sinh(\phi ae)$ $\sinh(\phi)$
η	$\frac{\tanh(\phi)}{\phi}$	$\frac{2I_1(\phi)}{\phi I_0(\phi)}$	$\frac{3}{\phi} \left[\frac{I_1(\phi)}{I_0(\phi)} - \frac{1}{\phi} \right]$

I_n is a modified Bessel function of order n

44

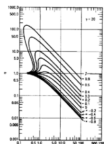
Falsified kinetics

- Cracking of cumene on $\text{SiO}_2\text{-Al}_2\text{O}_3$ catalyst



45

Non-isothermal pellet

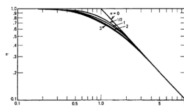


$$\gamma = \text{Arrhenius number} = \frac{E}{RT_s^2}$$

$$\beta = \frac{\Delta T_{\text{max}}}{T_s} = \frac{T_{\text{max}} - T_s}{T_s} = \frac{-\Delta H_{\text{rxn}} D_p C_{A0}}{k_s T_s}$$

46

For different orders



47

Problem

- A->B, 1st order, spherical particle ($d_p=2 \times 10^{-3}$ cm) coated with PL. The reactant concentration half way between external surface and center is $1/10^{\text{th}}$ of C_{A0} ($C_{A0}=0.001$ gmol/ m^3) $D_e=0.1$ cm²/sec. What is the reactant concentration at 3×10^{-4} cm from the external surface?
- What should be the diameter if $\eta=0.8$

48

Weisz-Prater criteria (C_{WP})

$$\eta \phi_1^2 = 3(\phi_1 \coth \phi_1 - 1)$$

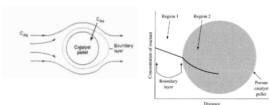
$$C_{WP} = \eta \phi_1^2 = \frac{\text{Observed (actual) reaction rate}}{\text{Reaction rate evaluated at } C_{A0}} \times \frac{\text{Reaction rate evaluated at } C_{A0}}{\text{A diffusion rate}}$$

$$C_{WP} = \eta \phi_1^2 = \frac{-r_A'(\text{obs}) \rho_p R^2}{D_e C_{A0}}$$

- > If $C_{WP} \ll 1 \Rightarrow$ no diffusional resistance, i.e., no concentration gradients in the pellet
- > If $C_{WP} \gg 1 \Rightarrow$ internal diffusion controls the reaction

49

External mass transfer resistance

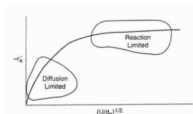


BTS Plant, Plant Campus

50

External mass transfer resistance

$$k_e = 0.6 \times \frac{D_{AB}^{1/3}}{v^{1/3}} \times \frac{L^{1/2}}{d_p^{1/2}}$$

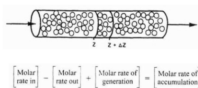


BTS Plant, Plant Campus

51

Mass transfer limited RXNs in packedbeds

- In MTL rxns, the surface reaction is rapid.
- Transfer of reactant from bulk to surface limits the overall reaction



BTS Plant, Plant Campus

52

Overall effectiveness factor

- Helps to analyze diffusion, flow and reaction in packed beds.
- Consider a situation where external and internal resistance to mass transfer to and within the pellet are of the same order of magnitude.



- At SS, the transport of reactant from the bulk to the external surface of the catalyst is equal to the net rate of reaction of the reactant within and on the pellet.
- Molar rate = (Molar flux) · (Ext. surface area) $\Rightarrow W_{dp} \cdot a_p \cdot \Delta V$

BTS Plant, Plant Campus

53

Overall effectiveness factor

- Molar rate of mass transfer to the surface, M_{s0} is equal to the rate of reaction (on and within)

$$M_{s0} = -r_s'' \text{ (external area + internal area)}$$

$$\text{External area} = \frac{\text{External area}}{\text{Reactor volume}} \times \text{Reactor volume} \\ = a_c \Delta F$$

$$\text{Internal area} = \frac{\text{Internal area}}{\frac{\text{Mass of catalyst}}{\rho_b} \times \frac{\text{Volume of catalyst}}{\text{Reactor volume}}} \times \text{Reactor volume} \\ = \frac{S_p}{\rho_b} \frac{P_b (1-\phi)}{\Delta F} \\ = S_p \phi_b \Delta F$$

BTS Pinar, Pinar Campus

54

Overall effectiveness factor

$$\Omega = \frac{\text{Actual overall rate of reaction}}{\text{Rate that would result if the entire surface were exposed to the bulk conditions, } C_{As0}, T_b}$$

$$\Omega = \frac{\eta}{1 + \eta k_p^2 S_p \rho_b / k_c \phi_c}$$

BTS Pinar, Pinar Campus

55

Mears criteria

$$\frac{-r_s'' \phi_b R_p}{k_c C_{As0}} < 0.15$$

- External resistance can be neglected

Where
 n = order of the reaction
 R = Cat. Particle radius
 ρ_b = bulk density
 C_{As0} = Cat. Conc.
 k_c = MTC, m/s

Another criteria

$$\left| \frac{-\Delta H_{Rx} (-r_s'') \rho_b R_p}{k T^2 R_p} \right| < 0.15$$

Where h = HTC between gas and pellet, KJ/m² . s.K

BTS Pinar, Pinar Campus

56

- A first order reaction is carried out using a spherical catalyst with a radius of 0.2 cm and effective diffusivity of 0.015 cm²/s. At 100 °C the intrinsic rate constant was measured to be 0.93 s⁻¹ with an activation energy of 20 kJ/mol.

- For a reactant concentration of 3.25 x10⁻² mol/liter, what is the rate of reaction at 100 °C?
- For the same reactant concentration, what is the rate of reaction at 150 °C? Assume that D_{eff} is constant, i.e independent of temperature.
- What is the apparent activation energy of the reaction?

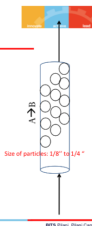
BTS Pinar, Pinar Campus

57

27-11-2023

Packed bed reactors

- Work horse of chemical/ petrochemical/ pharmaceutical
- Operated in two mode: Adiabatic & Non-adiabatic
- Steam reforming
- Catalytic reforming
- Isomerization
- Polymerization
- Ammonia synthesis and SO_2 synthesis



58

Configurations

- **Multi-bed**
Fluid stream goes into 1st reactor and then there may be some unit operation in between the PBRs
- **Reverse flow**
very useful to prevent catalyst deactivation. Eg. For exothermic reactions → The temp. increases along the axial direction → may lead to catalyst deactivation. After some time, switching will be done → indirectly preheat the feed
- **Counter flow reactor**
It is a two chamber reactor and an unique advantage is that, if the concentration is very lean, this kind of a configuration can actually be used to preheat the fluid which increases the reaction rate.
- **Radial flow reactor**
It is a two chamber reactor, the catalyst is placed between the annulus. The feed enters and flows radially (both the sides). It facilitates the operation under low pressure drop

BTS Plant, Plant Campus

59

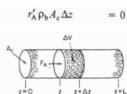
Mass transfer & reaction in packed bed

$$A_z W_{Az}|_z - A_z W_{Az}|_{z+\Delta z} + r'_A p_b A_c \Delta z = 0$$

$$-\frac{dW_{Az}}{dz} + r'_A p_b = 0$$

$$D_A \frac{d^2 C_{Ab}}{dz^2} - U \frac{dC_{Ab}}{dz} - \Omega p_b k'' S_x C_{Ab} = 0$$

$$X = 1 - \frac{C_{Ab}}{C_{Ab0}} = 1 - e^{-(\rho_b k'' S_x \Omega L / U)}$$



60

Problem

- Consider a 2nd order reaction $A \rightarrow B + 2C$ is carried out in a PBR. The gas is flowing at a temperature 260 °C, pressure 4.9 atm and with a superficial velocity of 4 m/s. Assuming external resistance is negligible. Find the length of the reactor to achieve 81% conversion.

Data: $k'' = 51 \times 10^3 \text{ m}^3/\text{mol.s}$; $D_p = 0.38 \text{ cm}$; $S_x = 410 \text{ m}^2/\text{g cat}$; density of the bed = $2.1 \times 10^3 \text{ g/m}^3$

BTS Plant, Plant Campus

61